Thermodynamics and Statistical Physics

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August 2025

Contents

1 Mathematical Foundations **2**
1.1 Some Useful Integrals . . . 2
1.2 Permutations and Combinations . . . 2
1.3 Probability Theory . . . 2

2 Classical Thermodynamics **3**
2.1 Introduction . . . 3
2.2 Zeroth and the first Law: . . . 4
2.3 Second Law: . . . 5
2.4 Free energy F and Legendre transformation . . . 7

3 Fundamentals of Statistical Mechanics **9**
3.1 Microcanonical Ensemble . . . 9
3.2 Math interlude: Counting Microstates . . . 10
3.3 The two level system . . . 11

4 Statistical Physics II **11**

5 Non-equilibrium statistical physics **12**
5.1 Introduction . . . 12
5.2 Mathematical Variables . . . 12
5.3 Brownian Motion . . . 13

1 Mathematical Foundations

1.1 Some Useful Integrals

The gaussian integral is defined by,

$$\int_{-\infty}^{\infty} e^{-ax^2}\ dx = \sqrt{\frac{\pi}{a}}$$

1.2 Permutations and Combinations

The most simplest problem is following: What are the total possible combinations of n distinct objects taken all at once ? The answer is $n!$. For context, this could be a problem such as given the letters A, B, C, what are the total possible combinations? $3! = 6$, ABC, ACB, CAB, CBA, BAC, BCA. Note that ABC and CBA are distinct items. Thus, in simple arrangement problem, the order/position of individual objects is relevant.

Now, in the same above problem, if the order was irrlevent, then the answer would be $\frac{n!}{n}$.

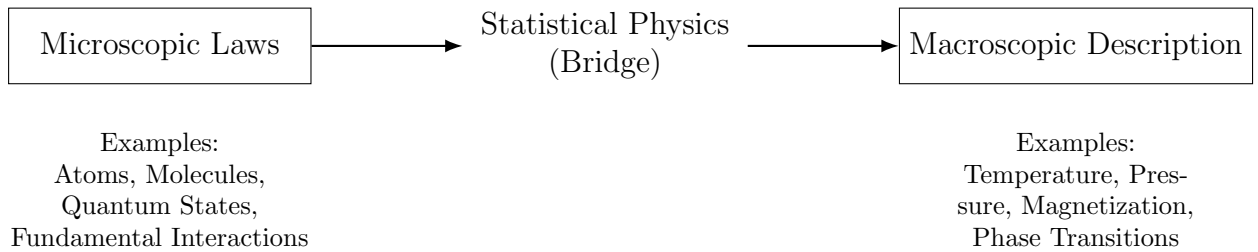
1.3 Probability Theory

We shall go through a brief and relevant discussion on probability theory.

2 Classical Thermodynamics

2.1 Introduction

Statistical physics is the bridge between the microscopic world and its laws, and the macroscopic description of the natural world.



Gas laws, Stefan Boltzmann law, Dulong-Petit law, are other examples of stuff that comes from more fundamental laws, either Newtonian or quantum mechanics mixed with a statistical dose.

$\star$) *New dictionary is invented in the process, for example it makes no sense to talk of temperature of an electron or atom, this quantity arises from statistical analysis of many many atoms and electrons.*

Thermodynamics itself is a phenomenological description of macroscopic systems at equilibrium without caring about the contents of the system. A more detailed study in statistical physics reveals that this phenomenological description arises from fundamental and common aspects of all systems.
Certain basic concepts from thermodynamics are intuitively familiar, like heating of two objects when rubbed against each other, heat transfer only taking place from hotter to colder body. These observations reveal an interesting aspect of thermodynamics, unlike fields such as classical mechanics or electromagnetism, thermodynamics is not tied to a specific physical domain, nor is it primarily concerned with calculating numerical values of physical quantities. Instead, it focuses on the general relationships between those quantities and the fundamental constraints that govern them.

Equilibrium may be defined as follows: an isolated system, when left undisturbed for a sufficiently long period of time, will evolve toward a state in which no further observable changes occur. This state is called equilibrium. In the special case where the variable of interest is the temperature, and it remains constant over time, the system is said to be in thermal equilibrium. Thermodynamics can be also defined as the determination of equilibrium states of a system. The equilibrium state properties are independent of the initial configuration of the system.

Put differently, thermodynamics concerns itself with the relationships between observable macroscopic phenomena in equilibrium state. An important property of an equilibrium state is that its characteristics depend only on the current external conditions, and not on the history or initial state of the system.
Just like in classical mechanics we have variables to define the state of particles; coordinates and momenta, similarly in thermodynamics, we have pairs like; Pressure and volume(for a fluid), Surface tension and area(for a film), tension and length (for a wire). There are other state variables as well that don't relate to mechanical work. All these variables are useful only for a system in equilibrium, a system whose properties don't change appreciably over sufficient time.

Importantly, in thermodynamics we are concerned only with macroscopic variables, not with the detailed behavior of individual atoms or molecules. This is advantageous because macroscopic variables typically evolve much more slowly than microscopic ones. For example, in a container of gas, the system as a whole can be characterized by quantities such as temperature, volume, and pressure, which change on relatively long timescales, whereas the positions and velocities of the individual molecules fluctuate rapidly due to random motion. Thermodynamics deals precisely with those states in which the macroscopic variables are constant or vary only extremely slowly.
The relationships among these macroscopic variables are expressed by the laws of thermodynamics, which

are phenomenological in nature and formulated from experimental observations rather than derived from first principles. A more fundamental explanation of these laws, based on the microscopic behavior of particles, is provided by statistical mechanics.

2.2 Zeroth and the first Law:

Zeroth law states that if A and B are in thermal equilibrium with a third body C, the they are also in thermal equilibrium with each other.

The first law is essentially the conservation of energy, it again is derived from countless experimental observations, it was deduced that the total energy of a system can be changed by with either work done or heat, that is,

$$\Delta E = Q + W$$

Change in the Internal Energy of the system = heat added + work done on the system

Thus, work and heat are just forms of energy transfer. More importantly, the fact that E is a state function, no matter what path we take for Q and W, as long as the initial and final states(p, V, N, T) are equal, the sum $Q + W$ will be same, thus E is a state function $E(p, V)$.
The standard convention in Physics is $Q > 0$ and $W > 0$ when heat is added to the system and work is done on the system. In chemistry, they use opposite convention for the work done.

Important: We can't write $E = Q + W$, this is wrong, what the first law tells us is that the energy change of a system is equal to heat absorbed by it plus the work done on it by the external environment.

Important: The first law in the above from is very general, it is applicable to any process, there is no restriction here. The process can be arbitrary — fast, turbulent, irreversible, whatever.

Now, we shift our focus to a *quasi-static* process, so that at every stage the energy change is very small, thus the system is effectively in equilibrium in each very small stage.

$$dE = d\bar{Q} + d\bar{W}$$

Where the bars signify that these are inexact differentials, what that means is that these correspond to path variables, both Heat Q and Work W depend on the path and not just on the initial and final stages of the system.

It is good to talk about work at this stage, from basic mechanics we know that work is just the force integral over position of the system. This is specifically called mechanical work. At each step of the quasi-static process, we have mechanical work given by $d\bar{W} = F\ dx$, this can be generalized to all types of work, not just mechanical, by the equation,

$$d\bar{W} = \sum_i J_i\ dx_i$$

Here x is a generalized displacement and J is the conjugate generalized force.
For pressure-volume work, we have,

$$dW = -P\ dV$$

A simple example for the first law would be a water inside a container. We can increase the temperature, hence the internal energy E by either heating the container, thus supplying Q, or by performing work by either an electric boiler inside the container or a mechanical pedal inside the container that would rotate.

System	Force		Displacement	
Wire	tension	F	length	L
Film	surface tension	$\mathcal{S}$	area	A
Fluid	pressure	$-P$	volume	V
Magnet	magnetic field	H	magnetization	M
Dielectric	electric field	E	polarization	P
Chemical reaction	chemical potential	μ	particle number	N

Figure 1: Different types of Generalized forces and displacements

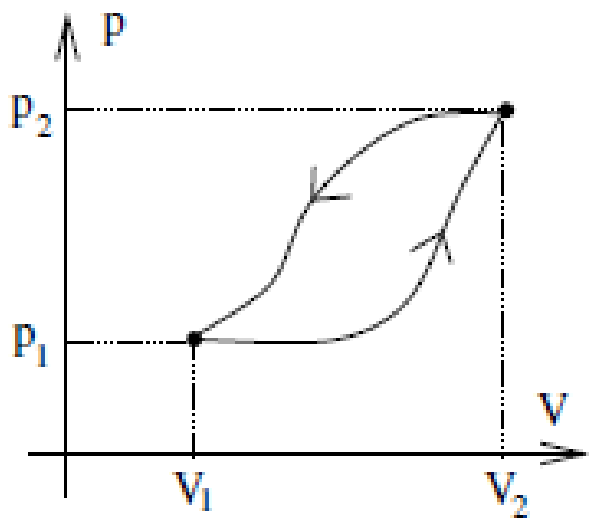


Figure 2:

2.3 Second Law:

To start the analysis of the second law, we begin with something called *reversible process*, these processes are quasi-static and plus there is no involvement of friction. Consider the process described in 2, starting from (p_1, V_1) go to (p_2, V_2) and then come back through the upper path. Since E is state function, we must have $\oint dE = 0$, but the total work done is not zero, we have,

$$\oint p\ dV = \oint d\bar{Q}$$

Thus there is an interconversion between heat and work, is the system loops around a path, going back to the initial position. If we go from lower path and then come back through upper path, the work done is negative, thus heat is negative and is released from the system. Going the other way, we see that heat is added to the system. Processes like these are interesting, they loop through their initial point and convert heat to work or vice versa.

Statement of the second law: There are two equivalent ways to state the second law:

1. Kelvin statement: No process is possible whose whole aim is to convert heat into work. (Concerns energy and its usage)

2. Clausius statement: No process is possible whose aim is to transfer heat from a colder to a hotter body. (Concerns the arrow of time)

These two statements are equivalent. Suppose for example we could violate the Kelvin statement, we build a process which uses heat from a hot source and entirely converts it into work, we can then use this work in a process like a fridge to cool down another source, then heat from this source can be deposited back into the hot source. But this violates the Clausius statement, since we just build a loop which dumps heat from colder to a hotter body.

To further illustrate the second law, we consider a special reversible cycle, called the Carnot engine or carnot cycle. It is a cycle that takes a system from a hot temperature T_H state A, firstly though isothermal expansion to state B, then through adiabatic expansion C having temperature T_C, then through isothermal compression to a state D and finally back to A through adiabatic compression. From A to B, the system absorbs some heat Q_H to maintain the hot temperature, and releases heat Q_C from C to D to maintain the cold temperature T_C. From the first law, the work done will be given by $Q_H - Q_C$, thus we define the efficiency of this process,

$$\eta = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

It can also be shown that,

$$\eta = 1 - \frac{T_H}{T_C}$$

As per the Kelvin statement, we can't have $\eta = 1$, we only have $\eta < 1$.

Carnot's theorem: Carnot's engine is the best we can do, that is of all the engines operating between two heat reservoirs, a reversible engine is the most efficient. Also, all reversible engines irrespective of their setup, operating between same temperatures, have same efficiency. Now we extract entropy from all this discussion,

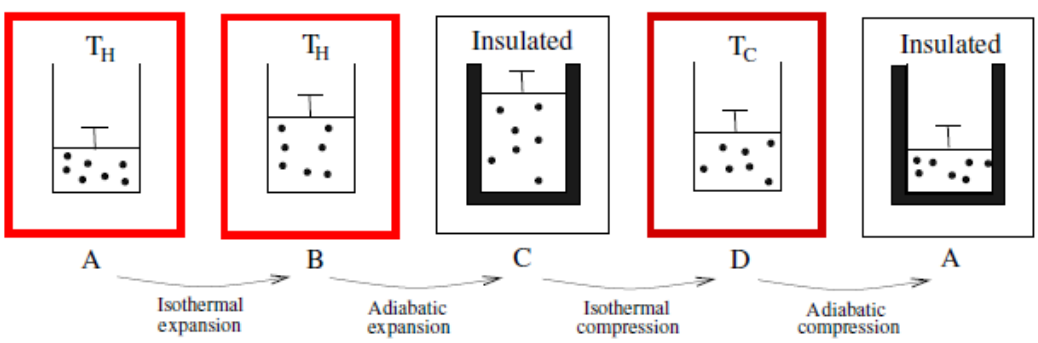


Figure 3: Carnot cycle

we notice that,

$$\frac{Q_C}{T_C} + \frac{Q_H}{T_H} = 0$$

This can be generalized to arbitrary number of steps,

$$\sum_i \frac{Q_i}{T_i} = 0$$

$$\oint \frac{d\bar{Q}}{T} = 0$$

It means that the quantity $\frac{d\bar{Q}}{T}$ must define a state variable. This is entropy, more precisely, we have,

$$S(A) = \int_0^A \frac{d\bar{Q}}{T}$$

Important: This thermodynamic entropy is only defined for equilibrium systems. For systems not in equilibrium we have to use other definitions such one from statistical mechanics.

Now, we look at an irreversible process, for this we can derive the following inequality, called the *Clausius inequality*,

$$\oint \frac{d\bar{Q}}{T} \leq 0$$

This leads us automatically to,

$$\Delta S \geq 0$$

Which is the second law of thermodynamics expressed in terms of entropy.
If the entropy of system is known as a function of extensive variables U, V and N, we have the *Fundamental relation*,

$$S = S(U, V, N)$$

All of thermodynamics can be derived in theory from the above fundamental relation.
Inverting the relation, we can write,

$$U = U(S, V, N)$$

$$dU = \left(\frac{\partial U}{\partial S}\right) dS + \left(\frac{\partial U}{\partial V}\right) dV + \left(\frac{\partial U}{\partial N}\right) dN$$

$$dU = TdS - PdV + \mu dN$$

Thus, we defined the temperature, pressure and electrochemical potential as,

$$T = \frac{\partial U}{\partial S} \quad P = -\frac{\partial U}{\partial V} \quad \mu = \frac{\partial U}{\partial N}$$

2.4 Free energy F and Legendre transformation

So far, we observed that using S and E as our central objects, we were able to write then in terms of extensive variables, which acted as mathematically independent variables. Then intensive variables were derived from the partial derivatives. This is kinda contrary to every day laboratory experience, where we are directly able to measure intensive variables like temperature and then deduce extensive variables like entropy or energy. Thus, we would like to modify the theory a bit, so as to swap the extensive and intensive variables. This can be done using Legendre transform, consider the function $U = U(S, V, N)$, we have,

$$T = \frac{\partial U}{\partial S}$$

We define,

$$F = U(S, V, N) - \frac{\partial U}{\partial S} S$$

$$F(T, V, N) = U(S, V, N) - TS$$

So, what we did was swapping S and T. We eliminated lS as an independent variable, even though we still have S on the right hand side, it can be in principle be expressed in terms of T, V and N.

Similarly, we can swap V and P, this will give us the enthalpy H,

$$H(S, P, N) = U(S, V, N) - \frac{\partial U}{\partial V} V$$

Now the simultaneous replacement of both entropy S and volume V, gives us Gibbs free energy or Gibbs potential,

$$G(T, P, N) = U(S, V, N) - TS + PV$$

Along with the total energy E and entropy S, we have another important variable, the Helmholtz free energy F,

$$F = E - TS$$

Free energy can be understand as "available" energy for work.

$$dF = dE - d(TS)$$

$$dF = d\bar{Q} + d\bar{W} - TdS - SdT$$

$$dF = -pdV - SdT$$

Thus, we have

$$p = -\frac{\partial F}{\partial V} \qquad S = -\frac{\partial F}{\partial T}$$

3 Fundamentals of Statistical Mechanics

3.1 Microcanonical Ensemble

The most easiest model to do statistical physics, is that of a Microcanonical ensemble. We consider an isolated system at fixed total energy E. The system has total N particles and volume V.

Then each configuration or the arrangement of the total energy E among the individual N particles constitutes a *Microstate*.
The *macrostate* is defined by the variables E, N and V. For each macrostate, we have many many microstates, whose number usually happens to be a absurdly large number.
Now we introduce the fundamental idea of a microcanonical ensemble: "All possible microstates are equally likely".

$$P = \frac{1}{\Omega(E)} \tag{1}$$

The **Entropy** S is defined by,

$$S = k_B \log(\Omega(E)) \tag{2}$$

This entropy is additive for two non-interacting systems. This is due to the fact that total number of microstates will be $\Omega = \Omega_1 \times \Omega_2$.
Second law: The second law of thermodynamics can be deduced from what we just learned; consider two non-interacting systems brought now in contact with each other. The total energy E is fixed $E = E_1 + E_2$. The total number of microstates will be,

$$\Omega(E_{\text{total}}) = \sum_{E_i} \Omega_1(E_i)\Omega_2(E_{\text{total}} - E_i)$$

$$\Omega(E_{\text{total}}) = \sum_{E_i} \exp\left(\frac{S_1(E_i)}{k_B} + \frac{S_2(E_{total} - E_i)}{k_B}\right)$$

$$S_{total} = k_B \log\left(\sum_{E_i} \exp\left(\frac{S_1(E_i)}{k_B} + \frac{S_2(E_{total} - E_i)}{k_B}\right)\right)$$

Usually, we have $S \propto N$, thus $\Omega(E_{\text{total}})$ is sum of exponential terms containing N, which is a big expression containing very large terms, for very large N, Ω_{total} will be dominated by the maximum values of Ω_1 and Ω_2.

$$\Omega(E_{\text{total}}) \approx \exp\left(\frac{S_1(E_*)}{k_B} + \frac{S_2(E_{total} - E_*)}{k_B}\right)$$

$$S(E_{total}) \approx S_1(E_*) + S_2(E_{total} - E_*)$$

Before the contact, the total entropy was,

$$S_{initial} = S_1(E_1) + S_2(E_{total} - E_1)$$

We can easily observe that,

$$S(E_{total}) = k_B \log \Omega(E_{total}) \geq S_1(E_1) + S_2(E_2)$$

Which is true, simple becase the original two system correspond to a subset of the total number of microstates. Thus, we get.

$$S_1(E_*) + S_2(E_{total} - E_*) \geq S_1(E_1) + S_2(E_2)$$

Now, if we consider the entire setup in a microcanonical ensemble, the most probable configuration occurs when the first system has energy E_*, the value that maximizes the total number of microstates of the combined

system — and therefore the total entropy. This directly reflects the second law of thermodynamics: when two systems are brought into contact, the total entropy tends to increase until it reaches a maximum. That maximum corresponds to the most probable configuration, since the largest possible number of microstates also represents the highest individual entropies for the subsystems.

Temperature: The temperature T is defined by,

$$\frac{1}{T} = \frac{\partial S}{\partial E}$$

We can see that this definition makes sense. We look at the concept that bodies in equilibrium have equal temperature. Consider two bodies($E = E_1 + E_2$) in contact with each other, from previous discussion we know that the system tends toward a configuration with maximum possible microstates or entropy, this is the equilibrium state.

$$\frac{\partial S}{\partial E_1} = \frac{\partial S_1(E_1)}{\partial E_1} + \frac{\partial S_2(E - E_1)}{\partial E_1} = 0$$

$$\frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2}$$

Thus, we can see that equilibrium condition corresponds to same T, which is expected for temperature.

Now during the transition to equilibrium, we expect energy to flow from hotter to colder object, this can be shown by writing,

$$\Delta S = \frac{\partial S_1}{\partial E_1}\Delta E_1 + \frac{\partial S_2}{\partial E_2}\Delta E_2$$

$$\Delta S = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\Delta E_1$$

We have already that $\Delta S \geq 0$, thus we obtain $\Delta E_1 > 0$ if $T_1 < T_2$ and $\Delta E_1 < 0$ if $T_1 > T_2$.

3.2 Math interlude: Counting Microstates

An important part of solving a microcanonical ensemble is being able to properly count the number of microstates. This concerns the concept of permutations and combination from math. We briefly go through these topics here.

The fundamental counting principles: If a task A is composed of n sub-tasks $A_1,\ A_2, A_3,\ ...A_n$, each realizable in $m_1,\ m_2,\ m_3,\ m_4,\ ...\ m_n$ different ways respectively. Then the total task A can be achieved in $m_1 \times m_2 \times\ m_3\ \times m_3\ \times\ ...\ \times m_n$ different ways. It is like we have a box containing n spaces for objects, the possible number of objects in each space is $m_1,\ m_2,\ m_3,\ m_4,\ ...\ m_n$, then the total ways to fill up the box will be $m_1 \times m_2 \times m_3 \times\ ...\ m_n$.
But if we have to events that can't happen simultaneously, like if we have three objects for first space and another two objects for the first space, then the total possible options is $3 + 2$, not the product.

Permutations(Arrangements): We look at the problem from a different perspective. The sum of the individual ways $m_1 + m_2 + m_3 + m_4 +\ ...\ + m_n = N$ is a constant number N. Also, now the individual ways are not independent, in the sense that once we use a way for the first sub task, we lose an option for the second task, and so the sequence continues to the next sub tasks.
Other way of understanding this is from the object and box perspective. We have total N objects($m_1, m_2, m_3, m_4,\ ...$) for the n individual spaces where we can put those objects, now we don't think in terms of the individual objects for each space, rather all the N objects are now put together in one basket.
Statement of the problem: *If we have n spaces and N distinct objects to be distributed among these space holders, then what is the number of maximum possible arrangements.*

$$P = N \times (N-1) \times (N-2)\ \times\ ...\ \times (N-n+1)$$

This can also be written as,

$$P(N,n) = \frac{N!}{(N-n)!}$$

Important points: The objects are all distinct, and the spaces occupy specified positions, that is order matters in the overall arrangement and each space can be occupied at max by just one particle.

For context; **if we have N number of particles and n energy states, such that all the particles are distinct and the particles occupy specific states only, then the total number of arrangements will be $P(N,n)$.**, which will be the number of microstates in this problem.
There are various equivalent ways of stating our original problem, another example is "What are the number of ordered arrangements of n objects selected from N distinct objects.

Permutations with repetition: Now, we look at another important problem that arises in arrangements, consider, that we have N objects and also N spaces, but now we have $n_1,\ n_2,\ n_3,\ ,...$ identical objects. For example, we are dealing with arrangements of the word "BALLOON", here L and O are repeated and we have $n_1 = n_2 = 2$. In general the possible number of arrangements here are,

$$P(N, n_1, n_2, ...) = \frac{N!}{n_1!\ n_2!\ n_3!\ \ ...}$$

3.3 The two level system

A simple application of what we have done so far, is the two level system.
Consider N number of non-interacting particles of total energy E(the energy is always fixed in microcanonical ensemble), such that each particle can be either in up state with energy $E_1 = \epsilon$ or downstate with energy $E_0 = 0$. The number of particles in up state is N_1(consequence of the fixed energy) and in lower state is N_0. Then the number of microstates, that is the number of possible arrangements,

$$\Omega(E) = \frac{N!}{N_1!N_0!} = \frac{N!}{N_1!(N-N_1)!}$$

This is the number of ways to arrange N distinguishable particles among N energy states, such that each state can be either state up or down. This problem corresponds to the case of **Permutations with repetition** that we discussed previously. Alternatively it corresponds to the case of **combinations**, that is permutations where order does not matter. Then, the entropy will be given by,

$$S = k_B \log\left(\frac{N!}{N_1!(N-N_1)!}\right)$$

4 Statistical Physics II

5 Non-equilibrium statistical physics

5.1 Introduction

Non-equilibrium systems show somewhat different behavior than equilibrium ones. At very small scales, that of atoms and molecules and in fields like biophysics, Nanoscience and chemistry, statistical fluctuations are too common to be completely ignored. In these domains, heat can sometimes jump from colder to hotter body and entropy can momentarily decrease. Also concepts like entropy don't automatically generalize to non equilibrium states.

There are two classes of non-equilibrium systems, (a) systems that start very near to a equilibrium state, but eventually return fully to an equilibrium state.

Apparent violations of second law?:

5.2 Mathematical Variables

In this section, we discuss some aspects of mathematical formalism.
In probability theory, each event is represented by a set A, if the set of possible events is given by set X, then the probability of event A is given by,

$$P(A) = \frac{n(A)}{n(X)}$$

Here $n(\ )$ is the number of elements of a set.
We are interested in figuring out the probability of event A, given that event B has happened, this is called conditional probability, defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Also, we have

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Intersection is symmetric operator, thus

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

This leads to the famous **bayes theorem**,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Now, if the entire sample space is composed of individual events A_i, such that,

$$\bigcap_i A_i = X$$

Then, we can write

$$P(B) = \sum_i P(B \cap A_i) = \sum_i P(B|A_i)P(A_i)$$

This is the **law of total probability**.
With this, we can rewrite the Bayes' theorem as,

$$P(A_k|B) = \frac{P(B|A_k)P(A_k)}{\sum_i P(B|A_i)P(A_i)}$$

For continuous variables, these expressions can be generalized as,

$$P(y) = \int P(y|x)P(x)\ dx$$

$$P(x|y) = \frac{P(y|x)P(x)}{\int P(y|x')P(x')\ dx'}$$

5.3 Brownian Motion

An important example of stochastic processes is the Brownian motion. We consider the following model to analyze the motion,

$$m\ddot{x} = -\gamma\dot{x} + f(t) - V'(x)$$

Which includes forces from friction, inter atomic interactions and external potential $V(x)$. Here $f(t)$ is in a way the fluctuating noise, such that $< f(t) >= 0$.

Case I: This equation is difficult to solve in the general case, we consider the case $V = 0$ and very large γ such that $m\ddot{x}$ term can be neglected. Thus we get,

$$\dot{x} = \frac{f(t)}{\gamma}$$

$$x(t) = x(0) + \frac{1}{\gamma}\int_0^t f(t')\ dt'$$

This leads to,

$$< x(t) >= x(0)$$

Which is pretty obvious, but what about the mean square displacement $< (x(t) - x(0))^2 >$? For this we need the following equation,

$$< f(t)f(t') > = \ 2D\gamma^2\delta(t - t')$$

This is the correlation function between $f(t)$ and $f(t')$. This gives us,

$$< (x(t) - x(0))^2 > = \ 6Dt$$

Case II: Now we consider the case when $m\ddot{x}$ can't be completely neglected, thus leading to,

$$m\ddot{x} + \gamma\dot{x} = f(t)$$

$$m\ddot{x}e^{\gamma t/m} + \gamma\dot{x}e^{\gamma t/m} = f(t)e^{\gamma t/m}$$

$$\frac{d}{dt}(\dot{x}e^{\gamma t/m}) = \frac{f(t)}{m}e^{\gamma t/m}$$

Which can be easily integrated to give,

$$\dot{x}(t) = \dot{x}(0)e^{-\gamma t/m} + \frac{e^{-\gamma t/m}}{m}\int_0^t f(t')e^{\gamma t'/m}\ dt'$$

The first term being the transient regime, goes to zero with time, thus for permanent regime we have

$$\dot{x}(t) = \frac{e^{-\gamma t/m}}{m}\int_0^t f(t')e^{\gamma t'/m}\ dt'$$

$$< \dot{x}(t) >= \dot{x}(0)$$